Charge radii of calcium isotopes within relativistic configuration-interaction density functional theory

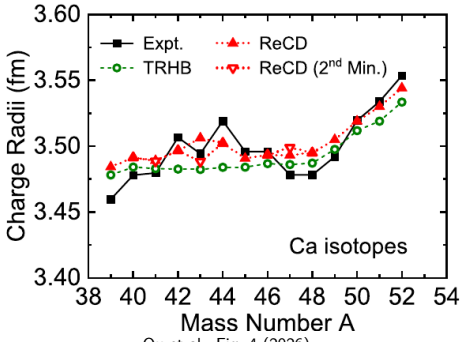
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Research background

Calcium charge radii are a multi-constraint puzzle



Qu et al., Fig. 4 (2026).

Endpoint constraint

"The anomaly in the charge radii along the Ca isotopic chain ...i.e., the almost equal values of charge radii in ^{40}Ca and ^{48}Ca and an inverted parabolic behavior between them, has been a longstanding challenge for nuclear theory in the past decades."

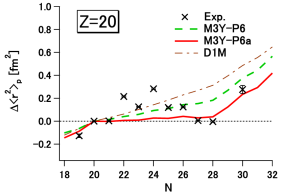
Huge radius of ^{52}Ca

An unexpectedly large radius was observed in neutron-rich calcium isotopes.

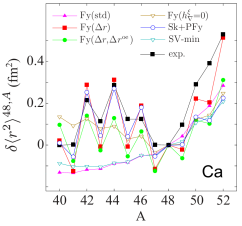
Odd-even staggering

Qu et al. report a pronounced OES in charge radii between ^{36}Ca and ^{48}Ca. Qu et al. (2026)

Mean-field remedies and their limitations



Nakada (2015).



Reinhard and Nazarewicz (2017).

Density-dependent spin-orbit term

"It is found that almost equal charge radii between ^{40}Ca and ^{48}Ca are reproduced ...owing to the density-dependence in the LS channel." Nakada (2015)

Fayans surface and pairing gradient terms

"By making the surface and pairing terms dependent on density gradients, the Fayans functional offers the superb simultaneous description of odd-even staggering effects in energies and charge radii." Reinhard and Nazarewicz (2017)

ReCD addresses the unresolved OES question

Ingredients

"In the ReCD theory, beyond-mean-field correlations are incorporated through rotational symmetry restoration and configuration mixing among various quasiparticle excited states."

"These two ingredients provide a microscopic treatment of rotational and vibrational correlations associated with zero-point motion, respectively." Qu et al. (2026)

Why it can

"An important advantage of this framework is that even-even, odd-A, and odd-odd nuclei can be treated on the same footing."

"This feature makes ReCD theory particularly suitable for exploring how beyond-mean-field correlations influence the odd-even staggering observed in the charge radii of calcium isotopes." Qu et al. (2026)

Research question

Can symmetry restoration plus qp configuration mixing reproduce the mid-shell enhancement and improve OES?

Theoretical framework

The nuclear charge radius

Paper Eq. (1)

The nuclear charge radius is evaluated as

$$r_c = \sqrt{\langle \hat{r}_p^2 \rangle + 0.64} \quad (\mathrm{fm}). \tag{1}$$

Here, the constant 0.64 accounts for the finite size of the proton, and $\sqrt{\langle \hat{r}_p^2 \rangle}$ denotes the root-mean-square radius of the point-proton density distribution. The corresponding mean-square radius is defined as

$$\langle \hat{r}_p^2 \rangle = \frac{1}{Z} \langle \Psi_I | \hat{r}_p^2 | \Psi_I \rangle = \frac{\langle \Psi_I | \hat{r}_p^2 | \Psi_I \rangle}{\langle \Psi_I | \hat{N}_p | \Psi_I \rangle}. \tag{2}$$

The ReCD many-body wave function

The nuclear many-body wave function $|\Psi_I\rangle$ is obtained within the framework of ReCD theory and reads

$$|\Psi_\sigma^{IM}\rangle = \sum_{K\kappa} F_{K\kappa}^{I\sigma} \hat{P}_{MK}^{I} |\Phi_\kappa\rangle. \tag{3}$$

where $\hat{P}_{MK}^{I}$ is the three-dimensional angular-momentum projection operator, and $\hat{P}_{MK}^{I}|\Phi_\kappa\rangle$ denotes the projected basis. The state $|\Phi_\kappa\rangle$ refers to an intrinsic state in the configuration space.

Configuration space for even-even nuclei

For even-even nuclei, the configuration space is constructed as

$$\begin{aligned}\{&|\Phi_0\rangle, \hat{\beta}^\dagger_{\pi_i}\hat{\beta}^\dagger_{\pi_j}|\Phi_0\rangle, \hat{\beta}^\dagger_{\nu_i}\hat{\beta}^\dagger_{\nu_j}|\Phi_0\rangle,\\ &\hat{\beta}^\dagger_{\pi_i}\hat{\beta}^\dagger_{\pi_j}\hat{\beta}^\dagger_{\pi_k}\hat{\beta}^\dagger_{\pi_l}|\Phi_0\rangle, \hat{\beta}^\dagger_{\nu_i}\hat{\beta}^\dagger_{\nu_j}\hat{\beta}^\dagger_{\nu_k}\hat{\beta}^\dagger_{\nu_l}|\Phi_0\rangle,\\ &\hat{\beta}^\dagger_{\pi_i}\hat{\beta}^\dagger_{\pi_j}\hat{\beta}^\dagger_{\nu_k}\hat{\beta}^\dagger_{\nu_l}|\Phi_0\rangle\}.\end{aligned} \tag{4}$$

Here, $\hat{\beta}^\dagger_\pi$ and $\hat{\beta}^\dagger_\nu$ represent the proton and neutron quasiparticle creation operators, respectively.

For the ground-state radii, the numerical calculation retains only the two-qp intrinsic states for even-even calcium isotopes; the four-qp terms in Eq. (4) are neglected.

The qp excitation-energy cutoff used to truncate the configuration space is

$$E_{\mathrm{cut}} = 3.5\ \mathrm{MeV}.$$

Configuration space for odd-N, even-Z nuclei

For odd-N and even-Z nuclei, it is given by

$$\left\{\hat{\beta}^{\dagger}_{\nu_0}|\Phi_0\rangle, \hat{\beta}^{\dagger}_{\nu_i}|\Phi_0\rangle, \hat{\beta}^{\dagger}_{\nu_i}\hat{\beta}^{\dagger}_{\pi_k}\hat{\beta}^{\dagger}_{\pi_l}|\Phi_0\rangle, \hat{\beta}^{\dagger}_{\nu_i}\hat{\beta}^{\dagger}_{\nu_k}\hat{\beta}^{\dagger}_{\nu_l}|\Phi_0\rangle\right\}. \tag{5}$$

For odd-N and even-Z nuclei, the quasineutron orbital ν_0 with the lowest qp energy is blocked during the self-consistent iteration of the TRHB equation in order to ensure the correct number parity.

For odd-A isotopes, only one-qp intrinsic states are retained; the three-qp terms in Eq. (5) are neglected.

The triaxial relativistic Hartree-Bogoliubov equation

All intrinsic states in Eqs. (4) and (5) are obtained by self-consistently solving the triaxial relativistic Hartree-Bogoliubov (TRHB) equation

$$\begin{pmatrix} \hat{h}_D - \lambda & \hat{\Delta} \\ -\hat{\Delta}^* & -\hat{h}_D^* + \lambda \end{pmatrix} \begin{pmatrix} U_k \\ V_k \end{pmatrix} = E_k \begin{pmatrix} U_k \\ V_k \end{pmatrix}. \tag{6}$$

Where U_k and V_k are the qp wavefunctions, λ is the Fermi energy, $\hat{h}_D$ is the single-particle Dirac Hamiltonian, and $\hat{\Delta}$ is the pairing field.

Both $\hat{H}^{\mathrm{RDFT}}$ and the TRHB equation are derived from the relativistic density functional PC–PK1. The TRHB equation is solved in a three-dimensional harmonic-oscillator basis with 10 major shells.

The Hill–Wheeler equation

The weight coefficients $F^{I\sigma}_{K\kappa}$ in Eq. (3) are determined by the following Hill–Wheeler equation

$$\sum_{K'\kappa'} \left\{ H^{I}_{K'\kappa',K\kappa} - E^{I\sigma} N^{I}_{K'\kappa',K\kappa} \right\} F^{I\sigma}_{K'\kappa'} = 0. \tag{7}$$

where

$$H^{I}_{K'\kappa',K\kappa} = \langle \Phi_{\kappa'} | \hat{H}' \hat{P}^{I}_{K'K} | \Phi_{\kappa} \rangle, \quad N^{I}_{K'\kappa',K\kappa} = \langle \Phi_{\kappa'} | \hat{P}^{I}_{K'K} | \Phi_{\kappa} \rangle$$

are respectively the energy kernel and the norm matrix.

The constrained Hamiltonian $\hat{H}'$

The Hamiltonian $\hat{H}'$ is written as

$$\hat{H}' = \hat{H} - \lambda_p(\hat{N}_p - Z) - \lambda_n(\hat{N}_n - N). \tag{8}$$

Here, $\hat{H} = \hat{H}^{\mathrm{RDFT}} + \hat{H}^{\mathrm{pair}}$ is the Hamiltonian of the system. Since the intrinsic states $|\Phi_\kappa\rangle$ are not eigenstates of the proton and neutron number operators, the ReCD many-body wave functions are generally not eigenstates of the nucleon number operators either. To restore the correct mean values of the nucleon numbers, we follow the standard prescription and introduce the additional constraint terms in Eq. (8).

The relativistic density-functional Hamiltonian

The $\hat{H}^{\mathrm{RDFT}}$ can be derived from a relativistic Lagrangian density through a Legendre transformation,

$$
\begin{aligned}
\hat{H}^{\mathrm{RDFT}} = \int \mathrm{d}\boldsymbol{x} \Big\{ & \bar{\psi}(-\mathrm{i}\boldsymbol{\gamma}\cdot\nabla + M)\psi \\
& + \frac{1}{2}\alpha_S(\bar{\psi}\psi)(\bar{\psi}\psi) + \frac{1}{2}\alpha_V(\bar{\psi}\gamma_\mu\psi)(\bar{\psi}\gamma^\mu\psi) + \frac{1}{2}\alpha_{TV}(\bar{\psi}\vec{\tau}\gamma_\mu\psi)\cdot(\bar{\psi}\vec{\tau}\gamma^\mu\psi) \\
& - \frac{1}{2}\delta_S\partial_0(\bar{\psi}\psi)\partial^0(\bar{\psi}\psi) - \frac{1}{2}\delta_V\partial_0(\bar{\psi}\gamma_\mu\psi)\partial^0(\bar{\psi}\gamma^\mu\psi) - \frac{1}{2}\delta_{TV}\partial_0(\bar{\psi}\vec{\tau}\gamma_\mu\psi)\cdot\partial^0(\bar{\psi}\vec{\tau}\gamma^\mu\psi) \\
& - \frac{1}{2}\delta_S\partial_i(\bar{\psi}\psi)\partial^i(\bar{\psi}\psi) - \frac{1}{2}\delta_V\partial_i(\bar{\psi}\gamma_\mu\psi)\partial^i(\bar{\psi}\gamma^\mu\psi) - \frac{1}{2}\delta_{TV}\partial_i(\bar{\psi}\vec{\tau}\gamma_\mu\psi)\cdot\partial^i(\bar{\psi}\vec{\tau}\gamma^\mu\psi) \\
& - \frac{1}{2}F^{0\nu}\dot{A}_\nu + \frac{1}{2}F^{i\nu}\partial_i A_\nu + e\bar{\psi}\frac{1-\tau_3}{2}\gamma^\mu A_\mu\psi \Big\}.
\end{aligned}
\tag{9}
$$

Pairing Hamiltonian and pairing force

Pairing correlations are treated by the pairing Hamiltonian

$$\hat{H}^{\rm pair} = \frac{1}{4}\sum_{\alpha\beta\gamma\delta}\langle\alpha\beta|\hat{V}^{\rm pair}|\gamma\delta\rangle\hat{c}^{\dagger}_{\alpha}\hat{c}^{\dagger}_{\beta}\hat{c}_{\delta}\hat{c}_{\gamma}. \tag{10}$$

For $\hat{V}^{\rm pair}$, we adopt a finite-range separable force,

$$\hat{V}^{\rm pair} = G\delta(\boldsymbol{R}-\boldsymbol{R}')P(\boldsymbol{r})P(\boldsymbol{r}')\frac{1}{2}(1-P^{\sigma}). \tag{11}$$

The standard pairing strength adopted in the calculations is

$$G = 728\ \mathrm{MeV\,fm^3}.$$

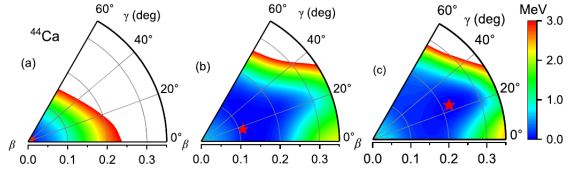
Final expression for the proton radius

Combining Eqs. (2) and (3), one obtains the final expression for the point-proton mean-square radius within the framework of ReCD theory,

$$\langle \hat{r}_p^2 \rangle = \frac{\sum\limits_{KK'\kappa\kappa'} F^{I\sigma *}_{K'\kappa'} F^{I\sigma}_{K\kappa} \langle \Phi_{\kappa'} | \hat{r}_p^2 \hat{P}^I_{K'K} | \Phi_\kappa \rangle}{\sum\limits_{KK'\kappa\kappa'} F^{I\sigma *}_{K'\kappa'} F^{I\sigma}_{K\kappa} \langle \Phi_{\kappa'} | \hat{N}_p \hat{P}^I_{K'K} | \Phi_\kappa \rangle}. \tag{12}$$

Results and discussion: potential-energy surfaces and deformation

^{44}Ca: how correlations move the energy minimum



Source: Qu et al., Fig. 1. PESs in the (β, γ) plane: (a) TRHB, (b) angular-momentum projection, and (c) full ReCD.

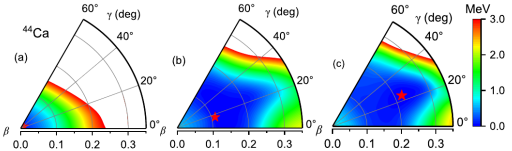
Three panels

(a) TRHB: minimum at $\beta = 0$; the surface is relatively stiff.

(b) Angular-momentum projection: minimum moves to $(0.11, 16°)$ and the surface becomes softer.

(c) ReCD: qp configuration mixing shifts the minimum further to $(0.21, 25°)$.

How projection and qp mixing shift ^{44}Ca deformation



Source: Qu et al., Fig. 1. PESs in the (β, γ) plane: (a) TRHB, (b) angular-momentum projection, and (c) full ReCD.

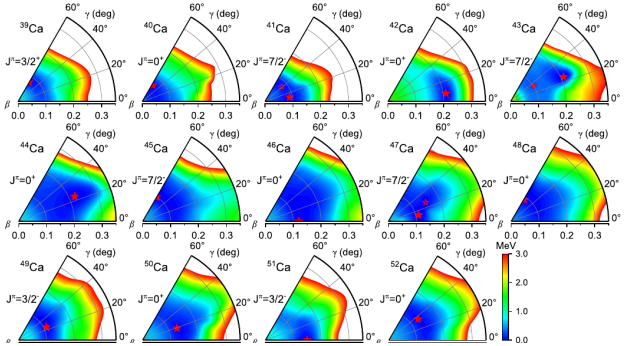
Paper's qualitative discussion

Beyond-mean-field correlations can induce dynamical shape fluctuations and drive the nucleus toward equilibrium deformations different from those predicted at the mean-field level.

The resulting shape changes are expected to have a direct impact on the charge radii according to Bohr's collective model.

$$\langle r^2 \rangle_c = \langle r^2 \rangle_{\beta=0} \left(1 + \frac{5}{4\pi}\beta^2\right).$$

ReCD yields diverse deformed minima along $^{39-52}$Ca



Source: Qu et al., Fig. 2. Filled stars: global minima; open stars: secondary local minima.

Classification

Oblate-dominated: ^{39,40,45,48}Ca.

Prolate-dominated: ^{46,51}Ca.

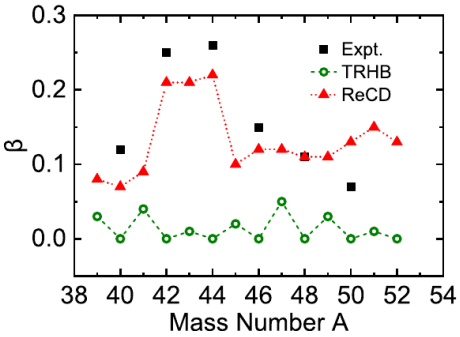
Triaxial: $^{41-44,47,49,50,52}$Ca.

Main contrast

TRHB predicts nearly spherical calcium isotopes, while ReCD gives a rich variety of equilibrium shapes.

This correlation-induced shape evolution appears throughout the studied calcium chain.

Deformation parameter β versus experiment



Source: Qu et al., Fig. 3. Deformation at PES minima; Expt is extracted from B(E2) data.

Paper's discussion

It is evident that the experimental β values effectively include dynamical correlations that are absent in static mean-field calculations.

When beyond-mean-field correlations are taken into account, the ReCD results reproduce the experimental β values reasonably well.

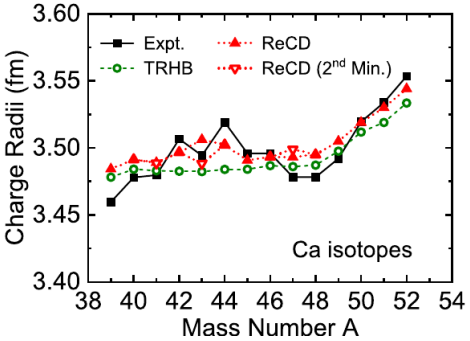
Geometric picture of radius

$$\langle r^2 \rangle_c = \langle r^2 \rangle_{\beta=0} \left(1 + \frac{5}{4\pi} \beta^2 \right).$$

Results and discussion: charge radii

Charge radii of calcium isotopes

Paper Fig. 4: ReCD versus TRHB and experiment



Source: Qu et al., Fig. 4.

TRHB

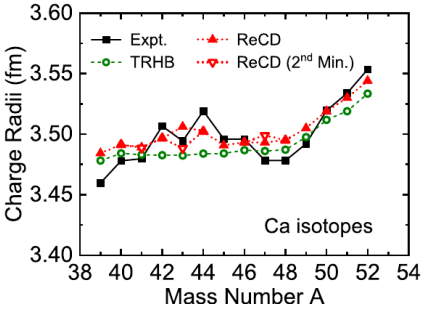
Reproduces the near-equal radii of ^{40}Ca and ^{48}Ca and the enhanced radius of ^{52}Ca, but underestimates $^{42–46}$Ca and fails to reproduce OES.

ReCD

Generally gives larger radii; preserves the ^{40}Ca/^{48}Ca near-degeneracy and ^{52}Ca enhancement; improves OES at ^{41}Ca–^{42}Ca and ^{44}Ca–^{45}Ca.

^{42}Ca–^{43}Ca: the remaining discrepancy

A secondary minimum gives a useful clue



Source: Qu et al., Fig. 4.

Paper's observation

The ReCD result does not fully reproduce the OES around ^{42}Ca–^{43}Ca, mainly because the predicted radius of ^{43}Ca is too large.

Secondary minimum

The radius associated with the secondary local minimum of ^{43}Ca is closer to experiment than the global-minimum result.

Results and discussion: what is explained?

Improved by ReCD

- equilibrium deformations move away from the nearly spherical TRHB picture;
- charge radii become generally larger;
- ^{40}Ca/^{48}Ca near-degeneracy is retained;
- the enhanced ^{52}Ca radius is retained;
- selected OES features are improved.

Limitations

- no exact particle-number projection;
- large-amplitude mixing between different shape points (β, γ);
- a complete explanation of the ^{42}Ca–^{43}Ca OES.

ReCD versus our MRCDFT+PROJ+GCM calculation

Aspect	Paper: ReCD calculation	Our MRCDFT+PROJ+GCM calculation
Intrinsic symmetry	Triaxial intrinsic states	Axially symmetric intrinsic states
Deformation coordinate	(β, γ)	β_2 only
Angular momentum	3D Angular-momentum projection $\hat{P}^I$	1D Angular-momentum projection $\hat{P}^I$
Particle number	No particle-number projection; average N, Z are constrained with Lagrange multipliers.	Explicit neutron- and proton-number projection
Configuration / shape mixing	Quasiparticle configuration mixing at fixed deformation point	Shape mixing among different deformation points through GCM
Correlation hierarchy	PES: TRHB $\rightarrow$ AMP $\rightarrow$ ReCD; r_c: TRHB versus full ReCD	CDFT $\rightarrow$ PROJ $\rightarrow$ PROJ+GCM
Charge-radius prescription	$r_c = \sqrt{\langle r_p^2 \rangle + 0.64\,\mathrm{fm}^2}$; finite proton-size correction	Two-body and center-of-mass corrections are included

Questions

THanks!

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